20. Number Sequences (Set B)

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Growing gap:

What comes next? 3, 5, 9, 15, 23, ? (the gap increases by 2 each time: +2, +4, +6, +8…)

Answer: The gaps are +2, +4, +6, +8, so the next gap is +10. 23 + 10 = **33**.

Example 2 — Fibonacci-style:

What comes next? 2, 3, 5, 8, 13, ? (each term is the sum of the two before it)

Answer: 8 + 13 = **21**.

Example 3 — Two-step rule:

A sequence follows the rule 'double the number, then subtract 1'. Starting at 2, what is the 3rd term? (2, 3, __)

Answer: Term 1 is 2. Term 2 = (2 × 2) − 1 = 3. Term 3 = (3 × 2) − 1 = **5**.

Example 4 — Multiple-choice, triangular numbers:

What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, ? A) 18 B) 20 C) 21 D) 24

Answer: Triangular numbers add one more each time: +2, +3, +4, +5, +6. 15 + 6 = 21. Answer: **C) 21**.

Example 5 — Select-all:

Select ALL that apply: which of these sequences follow a Fibonacci-style rule (each term is the sum of the two before it)? A) 1, 1, 2, 3, 5 B) 2, 4, 6, 8, 10 C) 2, 3, 5, 8, 13 D) 1, 2, 4, 8, 16 E) 4, 6, 10, 16, 26

Answer: A (1+1=2, 1+2=3, 2+3=5), C (2+3=5, 3+5=8, 5+8=13) and E (4+6=10, 6+10=16, 10+16=26) are all Fibonacci-style. B increases by a constant 2 (arithmetic), and D doubles each time (geometric). Answer: **A, C and E**.

Foundation

15 questions

1. What comes next? 5, 8, 11, 14, ?

2. What comes next? 80, 72, 64, 56, ?

3. What comes next? 4, 8, 16, 32, ? A) 48 B) 56 C) 64 D) 40

4. Select ALL that apply: which of these sequences increase by exactly 5 each time? A) 3, 8, 13, 18 B) 6, 10, 14, 18 C) 2, 7, 12, 17 D) 4, 9, 13, 18 E) 11, 16, 21, 26

5. What comes next? 9, 18, 27, 36, ?

6. What comes next? 95, 85, 75, 65, ? A) 50 B) 55 C) 60 D) 45

7. What comes next? 6, 11, 16, 21, ?

8. Fill in the missing number: 8, 16, __, 32, 40 (increasing by 8 each time)

9. What comes next? 2, 10, 50, 250, ? A) 1000 B) 1250 C) 1500 D) 750

10. Select ALL that apply: which of these are true about the sequence 5, 15, 45, 135? A) Each term is multiplied by 3. B) Each term increases by 10. C) The next term would be 405. D) The sequence is geometric. E) The sequence is arithmetic.

11. What comes next? 10, 15, 20, 25, ?

12. What comes next? 70, 63, 56, 49, ? A) 40 B) 41 C) 42 D) 43

13. What comes next? 3, 3, 6, 9, 15, ? (add the two previous terms)

14. Fill in the missing number: 5, 10, 20, __, 80 (each term doubles)

15. Select ALL that apply: which statements about the sequence 4, 9, 16, 25, 36 are true? A) These are square numbers, starting from 2^2. B) The next term is 49. C) Each term increases by a constant amount. D) The gap between terms increases each time. E) The sequence is decreasing.

Standard

20 questions

1. What comes next? 6, 9, 14, 21, 30, ? (the gap increases by 2 each time: +3, +5, +7, +9…)

2. What comes next? 160, 80, 40, 20, ?

3. What comes next? 4, 5, 9, 14, 23, ? (Fibonacci-style: add the two previous terms)

4. A sequence follows the rule 'double, then add 2'. Starting at 2, what is the 4th term? (2, 6, 14, __)

5. Select ALL that apply: which of these sequences follow a Fibonacci-style rule (each term is the sum of the two before it)? A) 2, 2, 4, 6, 10 B) 3, 6, 9, 12, 15 C) 1, 4, 5, 9, 14 D) 2, 4, 8, 16, 32 E) 5, 8, 13, 21, 34

6. What comes next? 4, 7, 13, 22, 34, ? (the gap increases by 3 each time: +3, +6, +9, +12…)

7. A sequence follows the two-step rule 'multiply by 2, then add 3'. Starting at 2, what are the first four terms?

8. What comes next in this sequence of triangular numbers? 15, 21, 28, 36, ?

9. In the sequence 2, 5, 10, 17, 26, ?, what is the rule linking consecutive terms, and what comes next?

10. Select ALL that apply: which of these sequences have a gap that increases by the same amount each time (i.e. the gaps themselves form an arithmetic sequence)? A) 2, 3, 5, 8, 12 (gaps 1,2,3,4) B) 4, 8, 12, 16, 20 (gaps all 4) C) 2, 4, 7, 11, 16 (gaps 2,3,4,5) D) 3, 6, 12, 24 (gaps 3,6,12) E) 1, 4, 8, 13, 19 (gaps 3,4,5,6)

11. What comes next? 480, 240, 120, 60, ?

12. A sequence starts 3, 4 and each new term is found by adding the previous two terms. What are the next three terms after 3, 4?

13. What comes next? 2, 4, 12, 48, 240, ? (hint: each term is multiplied by an increasing amount: ×2, ×3, ×4, ×5…)

14. Select ALL that apply: which of these are geometric sequences (each term multiplied by the same constant ratio)? A) 5, 10, 20, 40 B) 4, 8, 12, 16 C) 200, 100, 50, 25 D) 6, 18, 54, 162 E) 2, 5, 8, 11

15. A sequence follows the rule 'subtract 3, then multiply by 2'. Starting at 12, find the next two terms after 12.

16. What comes next? 3, 5, 9, 15, 23, ? A) 31 B) 32 C) 33 D) 34

17. A student claims the sequence 7, 14, 28, 56 is arithmetic. Explain why this is incorrect, and state what type of sequence it actually is.

18. Fill in the missing number: 7, 14, __, 56, 112 (each term doubles)

19. The triangular numbers are 1, 3, 6, 10, 15, 21, 28, 36… What is the 9th triangular number?

20. What comes next in this sequence, which alternates two simple operations: 3, 8, 5, 10, 7, ? (hint: alternately add 5 and subtract 3)

Hard

5 questions

1. What comes next? 3, 6, 18, 72, 360, ? (hint: each term is multiplied by an increasing amount: ×2, ×3, ×4, ×5…)

2. What comes next? 2, 3, 7, 16, 32, ? A) 56 B) 57 C) 58 D) 59

3. Select ALL that apply: which of these sequences follow a Fibonacci-style rule where each term is the sum of the two before it? A) 4, 7, 11, 18, 29 B) 2, 3, 6, 9, 15 C) 1, 5, 6, 11, 17 D) 10, 15, 20, 25, 30 E) 3, 4, 7, 11, 18

4. A sequence follows the rule ‘add 5, then double’. Starting at 1, find the next three terms after 1.

5. In the sequence 2, 6, 14, 30, 62, ?, each term follows the rule ‘double the previous term, then add 2’. What comes next?

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